

Step 0: Prerequisites & Tool Verification

Verify that your teammate's laptop has the required runtime environments installed prior to starting:

TOOL / DEPENDENCY	REQUIRED VERSION	VERIFICATION COMMAND (TERMINAL / POWERSHELL)
Java JDK	OpenJDK 17 or 21	<code>java -version</code>
Node.js	Node.js v18 LTS or v20 LTS	<code>node -v</code>
npm	v9+ or v10+	<code>npm -v</code>
MySQL Server	MySQL 8.0+ (Port 3306)	<code>mysql --version</code>

Step 1: Unpack Migration Package

1.1 Extract Zip Archives FILESYSTEM

Copy the **Smart_Nutrition_Migration/** folder to your local disk (e.g. `C:\Projects\Smart_Nutrition_Migration\`).

1. Extract `smart-nutrition-backend.zip` into `smart-nutrition-backend` directory.
2. Extract `smart-nutrition-frontend.zip` into `smart-nutrition-frontend` directory.

```
Smart_Nutrition_Migration/
├─ smart-nutrition-backend/      <-- (Extracted backend microservices)
├─ smart-nutrition-frontend/     <-- (Extracted frontend web app)
├─ smart_nutrition_db_full_dump.sql <-- (Full MySQL database dump)
└─ README_STARTUP.md           <-- (Setup reference)
```

Step 2: Database Import in MySQL

2.1 Create & Import Database MYSQL

Open PowerShell/Terminal in the `Smart_Nutrition_Migration` folder and execute:

```
# Create Database

mysql -u root -p -e "CREATE DATABASE IF NOT EXISTS smart_nutrition_db CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci;"

# Import Full Database Dump

mysql -u root -p smart_nutrition_db < smart_nutrition_db_full_dump.sql
```

⚠ **IMPORTANT:** The database name **MUST** be `smart_nutrition_db` so microservice database connections connect seamlessly without code changes.

Step 3: Configure Local Database Password & Secrets

3.1 Update secrets.properties CONFIGURATION

Open `smart-nutrition-backend/secrets.properties` and update `DB_PASSWORD` to match your local MySQL password:

```
GEMINI_API_KEY=AQ.Ab8RN6IGA4xM37jYMvqFe10calo_RckEbzmG5oW7WQbNtqPaaA
DB_USERNAME=root
DB_PASSWORD=YOUR_LOCAL_MYSQL_PASSWORD
```

Step 4: Install Frontend Dependencies

4.1 Run npm install FRONTEND

Open PowerShell in the frontend directory and install packages:

```
cd smart-nutrition-frontend
npm install
```

Step 5: Start Microservices (Strict Order)

Open **6 separate Terminal windows** and launch each microservice in sequence:

SEQ	MICROSERVICE	PORT	TERMINAL COMMAND
1	Eureka Discovery Server	8761	<code>cd smart-nutrition-backend/smart-nutrition-eureka-server && .\mvnw spring-boot:run</code>

SEQ	MICROSERVICE	PORT	TERMINAL COMMAND
2	API Gateway Service	8080	<code>cd smart-nutrition-backend/smart-nutrition-gateway && .\mvnw spring-boot:run</code>
3	Auth & Admin Service	8081	<code>cd smart-nutrition-backend/smart-nutrition-auth-service && .\mvnw spring-boot:run</code>
4	School & Student Service	8082	<code>cd smart-nutrition-backend/smart-nutrition-school-service && .\mvnw spring-boot:run</code>
5	Meal & AI Vision Service	8083	<code>cd smart-nutrition-backend/smart-nutrition-meal-service && .\mvnw spring-boot:run</code>
6	Messaging & Alerts Service	8084	<code>cd smart-nutrition-backend/smart-nutrition-messaging-service && .\mvnw spring-boot:run</code>

Step 6: Launch Web Application

6.1 Start Vite Dev Server FRONTEND

Open a 7th Terminal window:

```
cd smart-nutrition-frontend
npm run dev
```

Open web browser at: <http://localhost:5173>

Step 7: System Verification & Test Accounts

Log in using the pre-configured accounts to verify system operation:

ROLE	EMAIL ACCOUNT	PASSWORD	FEATURES TO TEST
Parent	dharun@gmail.com	sanjeet	Dashboard, Child Profiles, Lunchbox Presets, Lunch Scans, AI Assistant

ROLE	EMAIL ACCOUNT	PASSWORD	FEATURES TO TEST
Teacher	jothi@greenwood.edu	sanjeet	Class Overview, Student Roster, Nutrition Compliance, Parent Messages
Admin	admin@chewcheckers.com	sanjeet	User Accounts, School Settings, Audit Logs

Final Verification Checklist

- Eureka Service Registry:** <http://localhost:8761> shows all 5 services **UP**.
- Child Profiles & Presets:** Parent portal displays child **Sai** and presets (**Blue Bento Box**).
- Scanned Images:** Recent Lunch Scans display meal photos without broken image links.
- AI Assistant:** Clicking "What are high-protein lunchbox ideas?" streams formatted responses.
- Executive Summary:** Clicking "🚀 Ask AI Assistant for Detailed Summary" generates custom student insights.